5MMCS001W – 3D Interactive Media Development

Group Report – LEGO Pizza Delivery Simulator

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Concept and Design

We have created a 3D pizza delivery simulator in Unity, in LEGO style. The player is a pizza delivery driver in a small town made of LEGO. The primary goal is to pick up an order from the pizza shop, travel to the right customer, make the delivery, and receive points based on how quickly the delivery was completed. Once a delivery is completed, the player is given a new order, and the gameplay loop goes on.

We have chosen a pizza delivery game, as the goal is basic and immediately clear but still offers us a natural approach to combine the main areas requested by the module: 3D movement, camera control, interaction, triggers, collisions, animation, UI, navigation, scoring and physics. Instead of showing each instructional mechanic as a separate task to complete, we want to incorporate each tutorial mechanic into a single, unified interactive product.

The prototype is only a brief idea of the game, not the whole thing. The play area includes a pizza store, roads, a few houses or places to deliver to, environmental objects, a playable LEGO character, a waypoint system, a timer, scoring, and a simple user interface. This version offers us a complete gameplay loop, with space to expand in CW2.

Visual Style

We first drew inspiration for the visual style of our game from the typical basic LEGO kits found in stores. But we wanted the world to be more alive and engaging, not just like a static LEGO set. That's why we chose The LEGO Movie as one of our key visual inspirations. We want the setting to be colourful and use plastic-looking elements so that the pizza shop, roads, houses, vehicles and decorations all feel like they belong to this lifelike LEGO world. We also looked at LEGO City Undercover to see how a LEGO town can operate in a third-person game. The roads, buildings, cars, and open areas helped us think about how to construct our own delivery area and how the player will travel between key spots.

Gameplay References

We took Totally Reliable Delivery Service as inspiration for gameplay, as it revolves around collecting items and delivering them to different locations. We also looked at Crazy Taxi for its speedy objective system and the reward of getting to a destination swiftly. These ideas helped us make delivery speed the number one scoring criteria in our game. The core gameplay loop is Receive Order → Collect Pizza → Navigate to Customer → Deliver Pizza → Calculate Score → Receive Next Order.

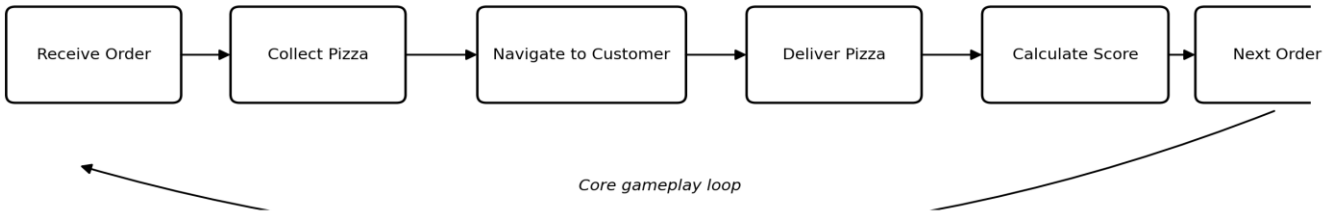


Figure 1. Core gameplay loop.

Design Diagrams

Level Layout / Diorama Plan

The level will be intentionally compact so that the player can understand the environment quickly while still having multiple delivery routes. The pizza shop acts as the main starting point, while several houses or buildings act as possible delivery destinations. Roads connect these areas and environmental props will be used to make the town feel lived in without making navigation confusing.

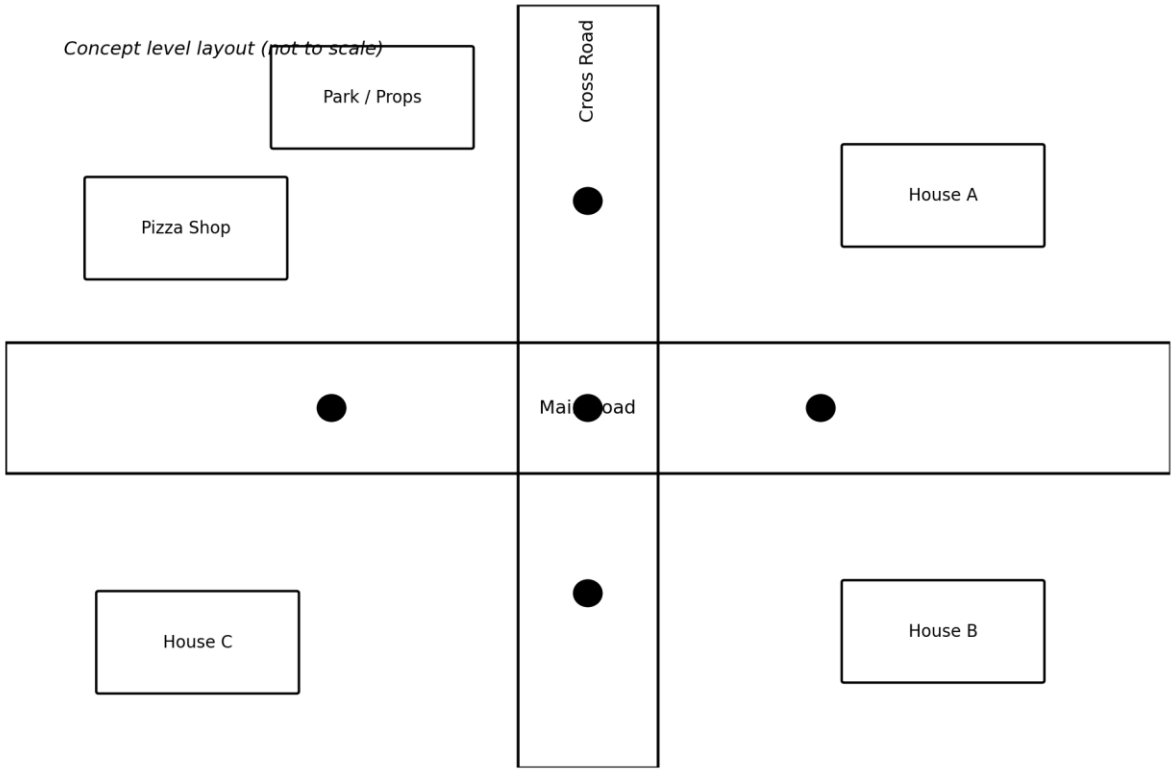
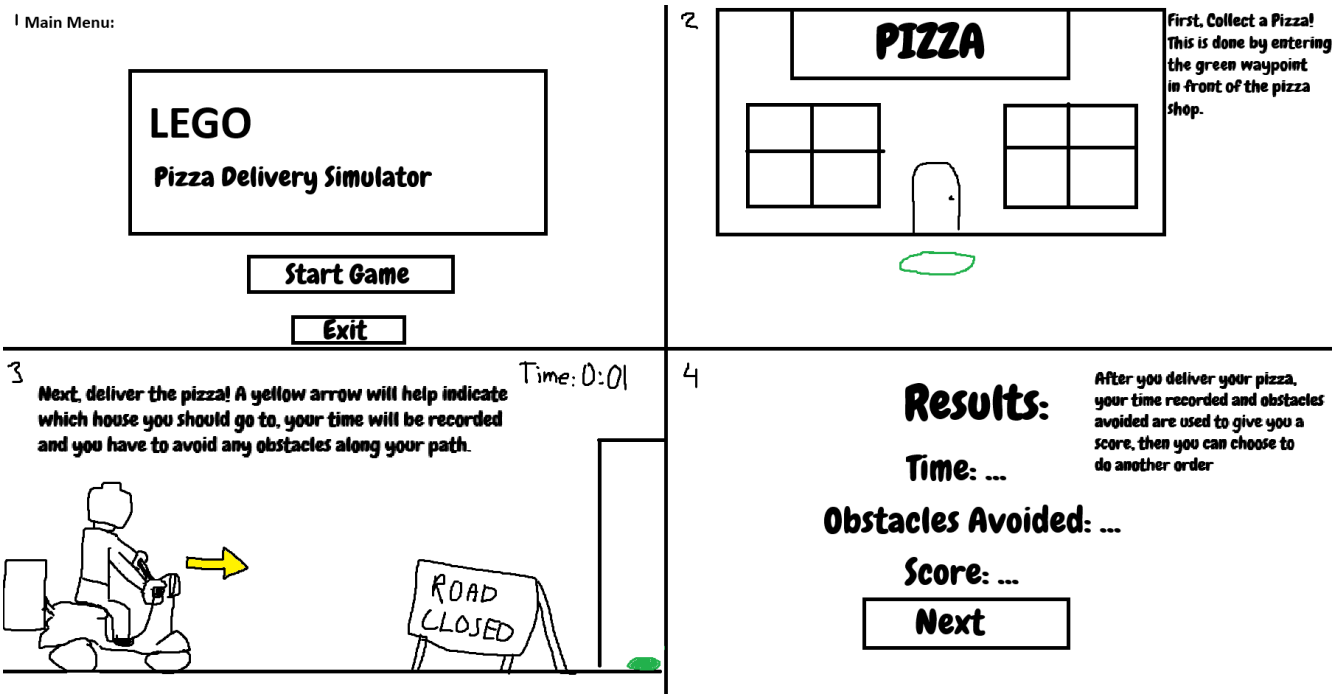


Figure 2. Early concept layout of the LEGO delivery area. Not to scale.

Storyboard



Flow Chart

The flow chart below shows the main gameplay sequence and the repeating structure of the experience. The purpose of this design is to keep the objective clear and reduce unnecessary complexity for the player.

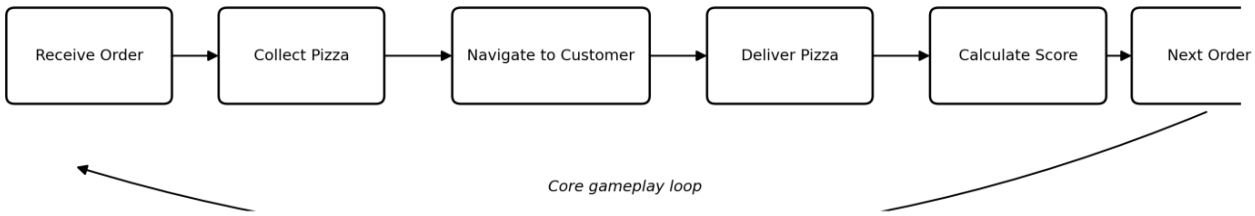


Figure 3. Delivery flow chart.

Use Case Diagram

The use case graphic displays the whole gaming system that we want to create in CW2. This includes pizza collection, delivery places, waypoints, timers, scoring, and complete delivery loop. CW1 focused

on the basic interactions such as player movement, camera controls, collisions, and tutorial-based systems. We were also able to add the view way point and HUD as extensions for project

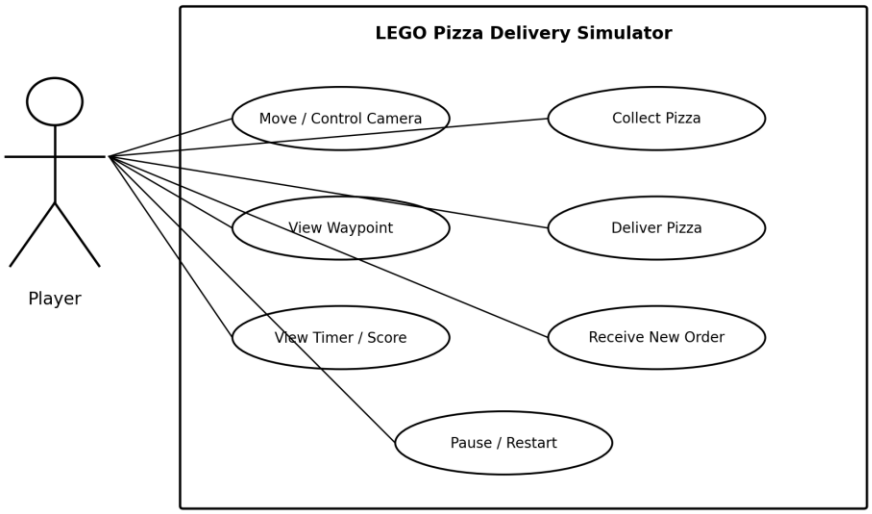


Figure 4. Basic use case diagram.

Domain / Class Diagram

The original idea was to break the project up into multiple scripts, each having a single principal duty. This would've been systems like a Game Manager handling the main delivery loop, and distinct scripts for pizza pickup, delivery spots, time, scoring, UI, player movement, and camera behaviors. Organizing the project in this way would make the code easier to follow, debug, and further develop during CW2. However, with the time available for CW1 we could only integrate the more basic scripts and instructional based systems. These were player movement, mouse input, Rigidbody physics, wall and collision behavior and character/controller scripts. These scripts are the foundation of the project and allow us to create a working interactive environment. More advanced scripts such as delivery management, scoring, timers and complete UI control that will be shown in the intended class diagram are the direction we want to take the project in CW2 rather than fully built systems in CW1.

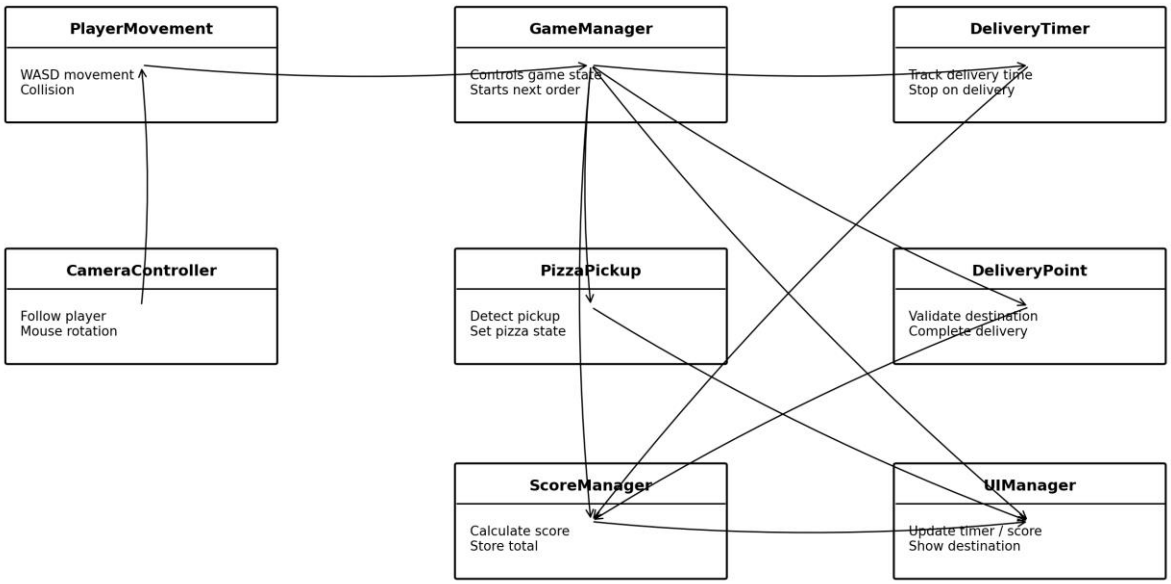


Figure 5. Planned relationship between the main Unity scripts.

Project Methodology

Our group workflow will be of the Agile and iterative nature. This means that the project will be built in smaller working stages, instead of trying to produce the complete game at once. For CW1 we want to complete and integrate the required lesson work into our LEGO pizza delivery scenario. When the required systems are working, we will add a few expansions to show how the project could develop. These extensions will be a delivery trigger point, a simple HUD, a timer, and a direction indicator. We will utilize GitHub to handle the shared Unity project and individual contributions. We will hold regular meetings to monitor progress, assign tasks, and decide which features to prioritize.

Individual Roles:

- Teo: Player movement, versioning and file management, presentations.
- Yuechen: Asset creation and sourcing, aesthetic design, camera control.
- Foyaz: Milestone updates, diagrams, meetings, administration, scheduling, physics controller.
- Malcom: Asset conversion, physics wall script.
- Everyone: Animation and implementation, animated movement, concept development, versioning.

Commercial Analysis

We built our pizza delivery simulator for casual gamers that like vibrant backgrounds, easy goals and timed obstacles. You can replay the game to try to improve your delivery times, explore alternative routes, and get better scores. A commercial version would need further testing, content and planning for costs, pricing, marketing and maintenance.

We would also examine the licenses of downloadable models, textures, audio, and animations. There may be restrictions on free assets too, so we need to check to see if they may be altered, used commercially or shared through GitHub. LEGO branding also needs consideration. Let us be aware that this is a student project. Before we could market the game, we would need permission to utilize protected LEGO branding and designs or replace them with our own original name, characters and materials. A disclaimer alone does not grant consent for us to use patented materials (LEGO Group, n.d.).

Navigation Based Framework

We will use both keyboard and mouse inputs in our navigation framework. WASD controls are used to maneuver the player across the area using the player's movement script, and the mouse controls the third-person camera using the mouse input script. The camera will be behind the player and will allow them to look around their surroundings while they are moving. Collision mechanisms will stop the player from walking through buildings, walls, and other solid things. Game play loop will be simple and flow as follows: player receives a delivery location, picks up a pizza at the restaurant, drives to the correct house, makes the delivery, and returns to pick up the next order. This will be ongoing throughout the game and after the game is finished, the credits screen will be displayed.

The main menu will be on the first screen, featuring options like Play and Controls. When the player starts the game, they are given simple instructions to learn the delivery objective and controls. You can press the Escape key to pause the game, and this will bring up choices such as controls or returning to the main menu. This keeps navigation easy and familiar to the player.

Interactivity

The main CW1 interactions will be through the tutorial systems and the mechanics already required by the module. This includes the demonstration of the player movement, mouse input, physics controller, and physics wall scripts and their applications into our project. The extensions that we would like to do are to add a delivery trigger point to the level. When the player hits this trigger, the game will know the player has reached the delivery area. The second would be a basic HUD which displays the time of the delivery and which house to deliver to. Third would be a basic waypoint which will point to the directions of the next deliver point and points of interest. The environment will also have the necessary tutorial interactions such as Rigid body objects, mouse-controlled objects, character controls, triggers, and slider-based interactions. Advanced technologies like multiple delivery locations, pizza pickup, order creation, and scoring will be developed during CW2.

Animations

The main LEGO character will be using the animations that the module requires, including the LEGO minifigure animation system. Basic movement animations, including idle and walking, may also be implemented when activated. More complicated animations like delivering pizza, passing over an order, customer reactions and ambient animations will be considered as future development. Due to our project's concept of delivering pizzas consisting of very limited actions causing, it requires few animations.

Lighting, Texturing and Shaders

The game's visual style will be inspired by The LEGO Movie, using bright colors, plastic-looking materials, and strong lighting to make the environment feel like a LEGO world. The LEGO characters, buildings, and props will use smooth and reflective materials to recreate the appearance of plastic LEGO pieces. Roads, pavements, and other environmental objects will use different colors and materials to keep the scene visually clear. The main aim is to create a clean and consistent visual style.

In the background, we will primarily employ simple Unity materials and textures, since these objects do not require complicated graphics. This will help keep the game operating smoothly, while still making the LEGO neighborhood look realistic and detailed. More essential items like the player, the pizza restaurant, and the delivery regions may use more reflective plastic-style materials to fit the LEGO look. We will try not to use too many complicated shaders, high-quality textures, or expensive lighting effects as these can degrade performance and drop the frame rate, especially when multiple objects are visible at once.

UI Effectiveness

We wanted to include a minimal HUD as an extension. The HUD will mostly just show a delivery timer and some basic navigational info. It will support gameplay by showing the player their objective and performance by helping them know their destination, how fast they should be going, and choosing better routes. The UI will use contrasting colors, clear text and simple icons so important information such as the timer, score and destination can be seen quickly. The UI will also provide immediate feedback by updating when a pizza is collected, a delivery is completed, or the player reaches the wrong location, however this will be developed later.

Use of AI

As part of our project we used AI to help us gain a better understanding of the CW1 coursework brief, requirements and marking criteria. This helped us to identify the key features and technical elements that are expected to be in our game and helped us to plan our development for us to gain higher marks. AI was used as a tool to help with understanding the assignment and organizing our priorities, not directly creating the content or technical systems of the game.

Conclusion

The key objective for CW1 was to finish the required lesson work and demonstrate this into one shared LEGO pizza delivery project. Basic player movement, mouse input, camera control, Rigidbody physics, character/controller scripts, collisions, trigger systems, and LEGO minifigure animation work are done. We also created the level and added the individual effort into the collective project in Unity. These elements are the basis of the project and illustrate the primary 3D interaction techniques taught in the lesson.

References

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7. LEGO Copyright and Trademark Policies: https://www.lego.com/en-us/legal/notices-and-policies/fair-play